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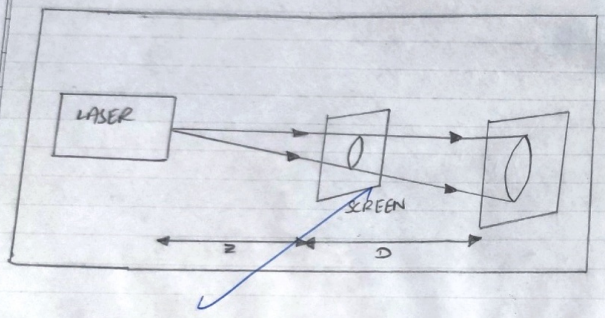
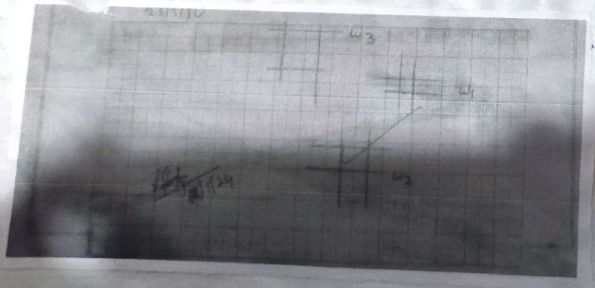
Experiment No. 1

Aim:- To measure the divergence of the laser beam

Apparatus:- He-Ne laser, stand, screen, measuring tape, graph, papers etc.





1 Expt. No. 1 Page No. 12/1/2023 Date

Experiment No. 1

⇒ **Aim:-** To measure the divergence of the laser beam.

⇒ **Apparatus:-** He-Ne laser, stand, screen, measuring tape, graph, papers etc.

⇒ **Theory:-** Divergence is defined as the spread of the laser beam i.e how much angle is subtended by the laser spot at the point of origin. It is measured in radian. In an optical resonator, if single-pass gain is more than cavity loss (including diffraction) then TEM$_{00}$ mode is developed whose intensity has distribution has emission shape in a direction transverse to the direction of propagation, of beam. A emission beam that has following property. If a beam has transmission transverse profile at one location, then it will egression traverse profile at the all locations else where (except when optical elements introduce a distortion, that is non-uniform across the wavefront of the beam). Such a egression beam can be characterized completely at any spatial location by defining both its "beam waist and radius of curvature of wavefront at specific location of the beam, moreover on unaltered. egression beam beam always has a minimum waist size. we at one location in space as shown in the figure.

In the figure, z-axis is chosen as direction of.

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Observation:- (i) Initial distance between Laser and screen is 2m.
(ii) Displacement of Laser D = 200 cm
(iii) Take five spot size.

S.No	Distance (cm)	spot size (cm)
1.	z = 200 cm	w_1 = 0.475 cm
2.	z = 400 cm	w_2 = 0.85 cm
3.	z = 600 cm	w_3 = 1.4 cm,

Calculations:- Angle of divergence is given by :-

$$\theta_0 = \frac{1}{D}\sqrt{\frac{w_3^2 - 2w_2^2 + w_1^2}{2}}$$

$$= \frac{1}{200}\sqrt{\frac{(0.475)^2 - 2(0.85)^2 + (1.4)^2}{2}}$$

$$= \frac{1}{200}\sqrt{\frac{0.225 - 1.445 + 1.96}{2}}$$

$$= \frac{1}{200}\sqrt{\frac{0.73}{2}} = \frac{1}{200}\sqrt{0.15} = \frac{1}{200} \times 0.44$$

θ_0 = 0.0022 rad
θ_0 = 2.2 × 10^{-3} rad.

Graph paper?

12/11/24

Result:- The angle of divergence of He-Ne laser is given by θ_0 = 0.0022 radian. since this angle is very small (= 1 milli radian), we conclude that laser beam is highly directional as compared to ordinary light.

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of propagation of laser, z-axis is chosen as direction of propagation of laser beam and origin is chosen at the point where waist size is minimum (say). since egaussian beam remains egaussian at all locations, thus the waist size of the beam at a distance z is given by the relation :

$$w^2(z) = w_0^2 + \theta_0^2 z^2 \quad \text{— (1)}$$

This result is a standard result and follows from property of egaussian beam. Here, w_0 = minimum waist size, θ_0 is the angle of divergence, w(z) is waist size at distance z.

To find angle of divergence θ_0, we set up three equations for this purpose, the waist or spot size (w) is measured at three arbitrary planes at z, z + D, z + 2D distance from the reference plan.

Let w_1, w_2, w_3 be spot size at distance z, z + D, z + 2D, from laser. then using equation (i), we have

$$w_1^2 = w_0^2 + \theta_0^2 z^2 \quad \text{— (2)}$$

$$w_2^2 = w_0^2 + \theta_0^2 (z + D)^2$$

$$w_2^2 = w_0^2 + \theta_0^2 z^2 + \theta_0^2 D^2 + 2z\theta_0^2 \quad \text{— (3)}$$

$$w_3^2 = w_0^2 + \theta_0^2 (z + 2D)^2$$

$$= w_0^2 + \theta_0^2 z^2 + \theta_0^2 (4D^2) + 4z D\theta_0^2 \quad \text{— (4)}$$

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From (2), (3), (4), we have

$$w_1^2 - 2w_2^2 + w_3^2 = 4\lambda^2 D^2$$

$$\lambda = \frac{1}{D}\sqrt{\frac{w_1^2 - 2w_2^2 + w_3^2}{2}} \quad - (5)$$

⇒ **Procedure** :- (i) Arrange the apparatus as shown in the figure.

(ii) Fix a graph paper on the screen and place it at some distance z (roughly 2m) from - laser. using a pencil & scale, draw horizontal and vertical tangents on both sides of spot so as to form a square shape of graph paper. using scale, measure mean diameter (distance between opposite sides) i.e waist size w_1.

(iii) Now displace screen in the direction of beam propagation by a known distance D (i.e total distance from laser is z + D) and measure spot size w_2 at the screen, as described in step (2). now displace screen further away by same value D (approximately 2m or more). so that new distance, of screen from laser is (z + 2D). Measure spot size w_3 at this position as described in step (ii).

(iv) Put values in formula and find laser divergence.

⇒ **Formula used** :- Let D = displacement of screen w_1, w_2, w_3 are spot size at distance z, z + D and z + 2D from laser, w_0 is minimum waist / spot size, then

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ngular divergence is given by

$$\theta_0 = \frac{1}{D}\sqrt{\frac{w_1^2 - 2w_i^2 + w_2^2}{2}}$$

⇒ **Precautions:-** (i) Experiment should be performed in dark room.

(ii) spot size should be measured accurately.

(iii) Screen should be placed normally to the path of laser beam.

(iv) laser light should not fall directly into the eyes of observer.

(v) spot size and other distances must be recorded in same units.

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⇒ **Result:-** The angle of divergence of He-Ne laser is given by $\theta_0 = 0.0022$ radian. since this angle is very small (= 1 milli radian), we conclude that laser beam is highly directional as compared to ordinary, light, source.

Aim :- To find dielectric constant of solid sample.

Apparatus :- A standard capacitor (50 pF), two gold plated, a dielectric sample, an AF oscillator, digital voltmeter, an inductor, cathode ray oscilloscope (CRO), screw gauge, vernier caliper, connecting wires, resistance,

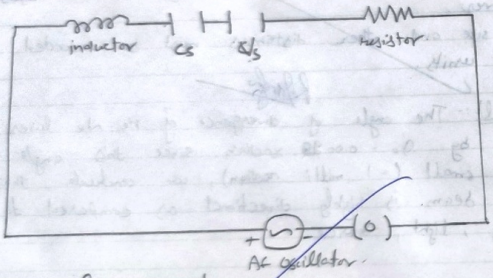

Fig :- Circuit

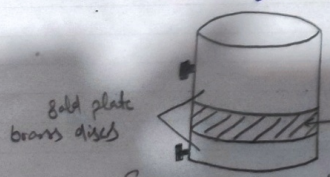

Fig :- Dielectric cell

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Experiment NO.2

Aim :- To find a dielectric constant of solid sample.

Apparatus required :- A standard capacitor (50 pF), two gold plated circular discs, a dielectric sample, an AF oscillator, digital voltmeter, an inductor, cathode ray oscillator, screw gauge, vernier calliper, connecting wires, resistance box etc.

Theory :- The circuit diagram for the experiment consists an indicator, a standard capacitor, a capacitor with dielectric slab, a suitable resistance, a source of variable frequency. Normaly AC from house hold power supply is not used because its frequency (50 Hz) is very low, capacitor offers high resistance ($X_{CS} = \frac{1}{C_S \omega}$) to low frequency.

Then frequency of ac source is checked suitably so that the passage of current through both capacitor is very easy. This is achieved by checking wave shape of AC using a CRO. The frequency is so adjusted so that a purely sinusoidal wave form of voltage / current exists in the circuit.

Suppose ω is angular frequency of ac passed.

$X_{CS} = \frac{1}{C_S \omega}$ = capacitive reactance of standard capacitor.

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#Observations:-

Capacity of standard capacitor : $C_s = 50$ pf
Least count of screw gauge :- 0.01 mm
Least count of vernier calliper :- 0.01 cm
voltage of standard capacitor = 2.28 V = V_s
voltage of dielectric cell = 2.43 V = V_d

Average diameter (D) = $\frac{0.0760 + 0.0756 + 0.0759}{3}$

= 0.0761 m

Average thickness (d) = $\frac{6.04 + 6.24 + 6.30}{3}$

= 0.006.22m

#Calculations:-

$C_d = \frac{V_s C_s}{V_d} = \frac{2.28 \times 50 \times 10^{-12} f}{2.43} = 4.6 \times 10^{-11} f$

$C_o = \frac{A\varepsilon_0}{d} = \frac{\pi D^2 \varepsilon_0}{4D} = \frac{3.14 \times (0.0761)^2 \times 8.854 \times 10^{-12}}{4 \times 0.006222}$

$= \frac{0.161 \times 10^{-12}}{0.024888} = 6.575 \times 10^{-12}$

$K = \frac{C_d}{C_o} = \frac{4.6 \times 10^{-11} f}{6.575 \times 10^{-12}} \Rightarrow 7.07$

#Result:-

$K = 7.07$

Robert 19/1/24

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D = Diameter of gold plated brass chain.
d = distance b/w gold plated brass discs
thickness of the dielectric sample.
K = dielectric constant of dielectric sample.
V_d = voltage across dielectric cell.
$A = \frac{\pi D^2}{4}$ = Area of each gold plated brass disc.

I_s = current passing through standard capacitor.
I_d = current passing through dielectric cell.

In series combination, current through all components remain same, hence:

$I_s = I_d$

$\frac{V_s}{X} = \frac{V_d}{X}$ $\Rightarrow V_s C_s = V_d C_d$

$C_d = \frac{V_s C_s}{V_d}$

Suppose dielectric slab is removed from gold plated brass discs such that separation between plate is still d. The capicity of capacitor becomes C_o which is given by $C_o = \frac{A\varepsilon_0}{d} = \frac{\pi D^2 \varepsilon_0}{4d}$

Let K be dielectric constant of the substances.
K = Capacity of capacitor with dielectric slab / capacity of capacitor without slab.

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Procedure :- (i) Connect AF CC greater to terminal of CRO and adjust calibration of generator so that purely sinusoidal wave is appeared.

(ii) Disconnect of generator from CRO and make connecting shown in fig.

(iii) Switch on the unit and using digital voltmeter. Find voltage across standard and dielectric coil.

(iv) Disconnect circuit and find thickness of dielectric slab and diameter of gold plate discs.

(v) Insert various value in standard formula to find dielectric slab.

(vi) Repeat experiment with different sample.

Formula used :-

$$K = \frac{4d\, V_s\, C_s}{\pi D^2 \varepsilon_0 V_d} \quad \text{OR} \quad K = \frac{C_d}{C_0}$$

Result :-

$$K = 7.07$$

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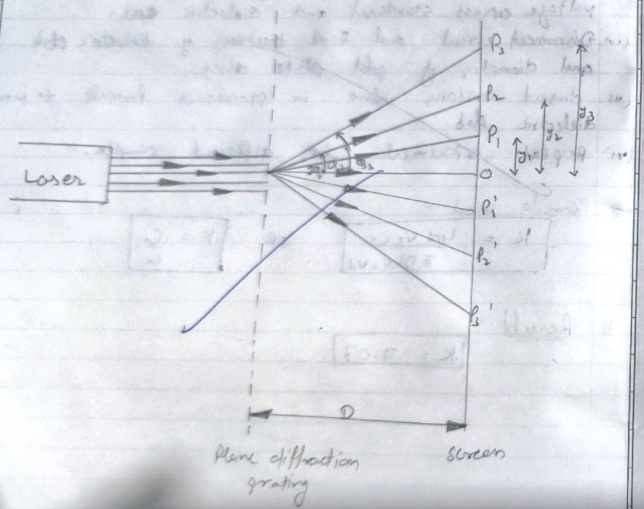
Aim :- To study diffraction using laser beam and this is to determine grating element.

Apparatus :- He-Ne laser, transmitter, grating element, measuring tap or steel scale, screen, graph paper



Expt. No. 3 Page No. 8 Date 27/2/2024

Aim :- To study diffraction using laser beam and this to determines grating element.

Apparatus :- A He-Ne laser, transmission grating, measuring tap or steel scale, screen graph paper etc.

Theory :- A diffraction grating is a extremely useful device to study diffraction. It consists a large number of slits placed side by side. All these slits lie in single plane. Due to this its also called plane diffraction. These slits are separated from each other opaque spaces when a wavefront is incident on a grating surface light is transmitted through a portion (transparent) and obstructed by opaque parts of grating. This causes diffraction of light and hence bright and dark fringes are obtained on screen. The intensity of bright fringes goes on decreasing in higher order.

Let a beam of rays is incident on a plane diffraction grating. The undiffracted rays are converged to its focussed by convex lens and these rays are from central maxima. However diffracted rays are maximum as well as minimum depending value of path difference.

Let, θ = angle of diffraction

a = AB = width of "transparent path"

b = BC = width of opaque part

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Observation and Calculation :-

Wavelength of laser beam (λ) :- 6.5×10^{-5} cm

Distance between diffraction grating and screen (D) :- 22.8 cm

Sr.No	Distance b/w corresponding maxima	y_n	Angle of Diffraction
1	$P_1 P_1' = 3.2$ cm	$y_1 = \frac{3.2}{2} = 1.6$ cm	$\theta_1 = \tan^{-1}\left(\frac{1.6}{22.8}\right) = 4.004$
2	$P_2 P_2' = 6.3$ cm	$y_2 = \frac{6.2}{2} = 3.1$ cm	$\theta_2 = \tan^{-1}\left(\frac{3.1}{22.8}\right) = 7.8$
3	$P_3 P_3' = 9.4$ cm	$y_3 = \frac{9.4}{2} = 4.7$ cm	$\theta_3 = \tan^{-1}\left(\frac{4.7}{22.8}\right) = 11.3$

$$d = \frac{n\lambda}{\sin\theta_n}$$

$d_1 = 1 \times 6.5 \times 10^{-5} / 0.06 = 103.5 \times 10^{-5}$

$d_2 = 2 \times 6.5 \times 10^{-5} / 0.13 = 100 \times 10^{-5}$

$d_3 = 3 \times 6.5 \times 10^{-5} / 0.19 = 102.6 \times 10^{-5}$

Mean value of grating element is $\frac{d_1 + d_2 + d_3}{3} = 102 \times 10^{-5}$

Result :- Grating element is 102×10^{-5}

Actual value :- 1 mm = 100 slits

10^{-1} cm = 100 d

$d = 10^{-3}$ or 100×10^{-5}

Percentage error :- $\frac{\text{observed value} - \text{actual value}}{\text{actual value}} \times 100$

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Let $d = A + B$

d is called slit width or and grating element is equal to sum of width and transparent and opaque part equal to

In $\triangle ACD$ we have

$$\frac{CD}{AC} = \sin\theta$$

$CD = AC\sin\theta = (AB + AC)\sin\theta$

$CD = d\sin\theta$

Point P on screen will be a secondary maxima when path difference CD is integeral multiple of wavelength of light i.e :- $CD = n\lambda$

$d\sin\theta = n\lambda$ ⊕

The equation called grating eq^n and gives the condition for formation of secondary maxima from diffraction grating. Here θ is replaced by θ_n, which simply show that angle of diffraction is different for different orders.

Let, there are N slit width in a unit length of a diffraction grating. d is related to it find relation between N and D we note that one slit width in unit distance $= \frac{1}{d}$

$$N = \frac{1}{d}$$

we can find N using (iii) and (iv) equation as follows

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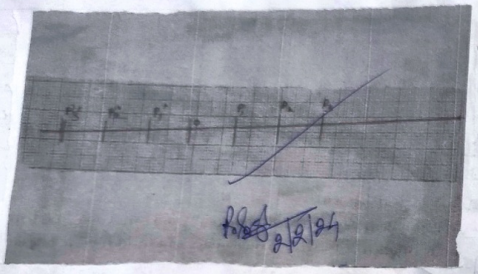
$= \frac{10^{-2} \times 10^{-5} + 100 \times 10^{-5}}{100 \times 10^{-5}} \times 100$

$= 2\%$



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Diffraction of light waves cannot be observed only if slit width d is suitable chosen $\Rightarrow \frac{n\lambda}{d} \leq 1$

$d \geq n\lambda$

Since maximum value of n is one for diffraction to be observed. Hence $d_{min} \geq \lambda$

Thus diffraction can be observed only if size of slits is comparable to wavelength of light used.

formula used :-

let, d = grating element

θ_n = angle of diffraction

n = order of diffraction

λ = wavelength of laser beam.

$d = \frac{n\lambda}{\sin\theta_n}$ $\quad \theta_n = \tan^{-1}\left(\frac{y_n}{D}\right)$

where, y_n is mean distance between nth maxima and central maxima.

D = distance between diffraction grating and screen.

Procedure :-

1. fix the apparatus.
2. fix a graph paper on screen and place it at appreciable distance D from grating so that distinct maxima are seen.

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Mark the centre of central spot (brightest) and first three maxima on both sides of central maxima.

4) using a scale and find distance between first maxima on either side of central spot. Then mean distance of first maximum from centre of screen is half of this separation i.e

$$y_1 = \frac{|PP'|}{2}$$

find angle of diffraction using $\tan\theta_1 = \frac{y}{D}$ then use this value in grating equation to find d grating element

5) Repeat the procedure of step 4) for 2nd and 3rd maxima. Then find value of mean N.

Precautions:-

1) Experiment must perform in dark room.
2) Laser light should not fall on eyes of observer directly.
3) All lengths should be measured in same units.
4) Distance between spots should be measured accurately.

Source of Error:-

(i) Distance between spot was not measured accurately.
(ii) There can be some calculation error.
(iii) All the distances may not be taken in same unit.

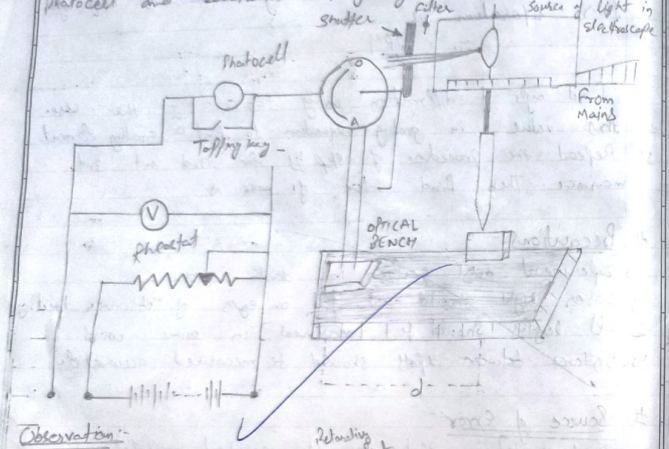
* Result:- Grating element of given diffraction is $=102 \times 10^{-5}$ cm.

Teacher's Signature

Experiment - 4

Aim :- To study planck's constant and work function by using photoelectric effect.

Apparatus :- photo tube, 0.2 DC supply, yellow filter, 100 W filament lamp, power supply, 2 DPM sensitivity, mA coaxial cable to join photocell and electrometer, range of colour filters, source of light in electrometer.



Observation :-

Reading for

S.NO	λ (nm)	ν (×10^14) (Hz)	V_s (V)	V_s (×10^14)	ν^2 (×10^28)
1	620	4.83	0.1	0.483	23.32
2	580	5.17	0.12	0.6204	26.72
3	520	5.76	0.36	2.073	33.17
4	490	6.12	0.69	4.4202	40.70
	Σ =	22.14	1.37	7.57	123.21

Expt. No. 4 Page No. 12 Date

Experiment - 4

Aim :- To find planck's constant and work function by photoelectric effect.

Apparatus Required: vacuum photocell with potassium surface mounted on lighted box, 0.2 V DC supply, 1 DPM sensitivity, mA coaxial to join photocell and electrometer, range of colour filter, 100 W filament lamp.

Theory :- It was observed that most metal under influences of radiation emits electrons, this phenomena known as photoelectric effect. That emission process depends upon critical frequency unable to emits electron, this freq known as work function.

The excess energy of photon appears as kinetic energy of the electron.

$$h\nu = \frac{1}{2}mv^2 + \omega$$

$$\boxed{\frac{1}{2}mv^2 = h\nu - \omega}$$

If we apply a retarding pot V_s so as to stop the photoelectron completely, it is known as stopping pot. at any instant :- $\frac{1}{2}mv^2 = eV_s$

$$eV_s = h\nu - h\nu_0$$

$$\boxed{V_s = \frac{h}{e}(\nu - \nu_0)}$$

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$$m = \frac{\sum n \sum y - n \sum ny}{(\sum n)^2 - n(\sum n^2)} = \frac{22.1 \times 10^{14} \times 23.8 - 6 \times 7.54 \times 10^{14}}{(23.8 \times 10^{14})^2 - 6 \times (98.331 \times 10^{28})}$$

$$= \frac{(528.1178 - 30.948) 10^{14}}{(490.12 - 485.172) 10^{28}} = \frac{2.6}{5.54} = 0.39 \times 10^{-14}$$

$$h = mxe = 0.39 \times 10^{-14} \times 1.6 \times 10^{-19}$$

$$= 0.624 \times 10^{-33} \Rightarrow \boxed{6.24 \times 10^{-34} \text{ Js}}$$

$$C = \frac{\sum xy - n \bar{x} \bar{y}}{\sum n} = \frac{(7.54 - 0.39 \times 23.83) \times 10^{14}}{22.14 \times 10^{14}}$$

$$C = -1.84$$

work function (ω) $= c \times e \Rightarrow (-1.84 \times e)$ J

$= +1.84 e$ V

$\omega = 1.84$ V

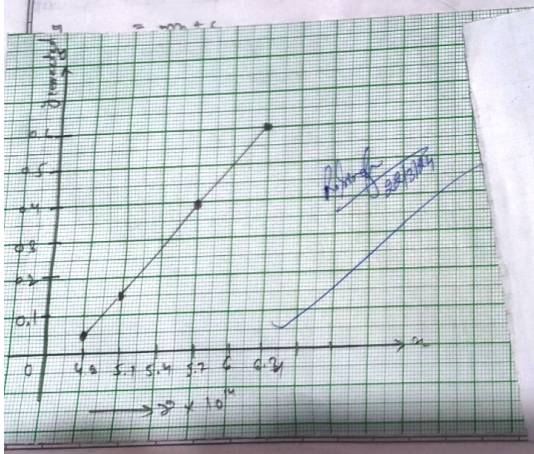
Rishabh 22/3/24

$= mx + c$

constant is

1.84 eV

(Cesium)



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If we plot a graph h gives slope of graph considering frequency of x-axis and $\frac{e}{10}$ stopping pot on y-axis.

$\frac{h}{e} = m$

$\boxed{h = mxe}$

Procedure:- 1. Set the equipment.
2. Insert the colour filter, set light intensity at strong light. voltage direction at '-' stopping mode switch at current display.
3. Adjust to de-accelerating voltage to 0V and set current display, decrease de-accelerating voltage to decrease to photo-current to 0, when photocurrent becomes 0.
4. Note down the de-accelerating voltage (V0), ie stopping potential.
5. Repeat above process three times.

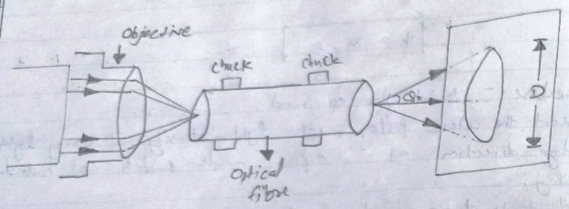
Precautions:- 1. The instrument should be operated in a dry, cool, and indoor space.
2. If there is dust on the phototube, colour filter, clean it by using absorbent cotton with a few drops of alcohol.
3. The colour filter should be stored in dry and dust proof environment.

Result:- 1. The value of Planck constant is 6.24×10^{-34} Js.
2. work function is -1.84 eV. (Cesium).

Teacher's Signature

Aim :- To determine Numerical Aperture of an optical fibre.

Apparatus :- He-Ne laser, optical fibre, screen, graph paper, fibre optics chuck, measuring tape etc.



Observations :-

(c) distance b/w Screen and optical fibre(L) = 0.8 cm

(d) diameter of spot = $\frac{D_1 + D_2}{2} = \frac{0.8 + 0.8}{2} \Rightarrow 1.2$ cm

$\theta_0 = \tan^{-1}\left(\frac{D}{2L}\right)$

$= \tan^{-1}\left(\frac{1.2}{2 \times 0.8}\right)$

$= \tan^{-1}(0.75) = 36.86°$

$\sin\theta_0 = \sin(36.86) = 0.599.$

Result :- NA is 0.599

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Expt. No. 5 Page No. 14 Date

Experiment-5

Aim :- To determine Numerical Aperture of an optical fibre.

Apparatus :- He-Ne laser, optical fibre, screen, graph paper, measuring tape, fibre optic chuck.

Theory :- Numerical Aperture is a basic characteristics of optical fibre. It can be thought as representing the size or degree of input acceptance cone.

mathematically, it is defined as the sine of angle of acceptance.

$$NA = \sin\theta = \sqrt{n_1^2 - n_2^2}$$

To measure numerical Aperture experimentally, we consider light comes out of optical fibre makes same acceptance cone at output end. Let L be the distance b/w screen and optical fibre. Let D is be diameter of the surface circular spot then :- we can write :- $\tan\theta_0 = \frac{D}{2L}$

$$\theta_0 = \tan^{-1}\left(\frac{D}{2L}\right)$$

after finding $\sin\theta_0$, find $\sin\theta$.

Procedure :- i) Arrange the apparatus, mount both ends of optical fibre on the fibre optics chucks.

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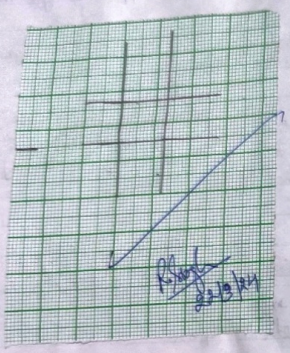

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(ii) Place the screen at some distances from the output end of the fibre.

(iii) Now, move the source towards and away from output end of optical fibre, such that a circular spot is formed on the screen.

(iv) Measure the distance between output end and and screen. Also measure the diameter of the circular spot of formed on the screen.

Precautions :- (i) ~~spot~~ formed on graph paper etc. shown be sharply defined.

(ii) ~~Experiment~~ must have performed in dark room.

(iii) coupling of light to optical fibre should be accurate.

Result :- Numerical Aperture is 0.62.

Teacher's Signature

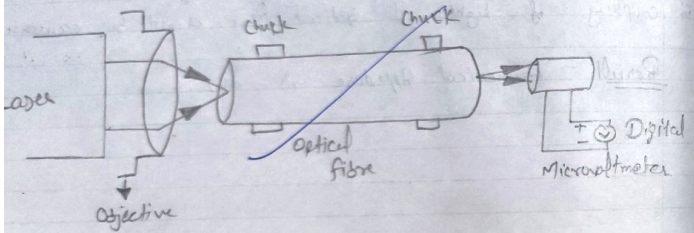
Experiment - 6

Aim :- To determine attenuation and propagation losses in optical fibre

Apparatus Required :- He-Ne laser, 20× microscopic objective, fibre chuck, two optical fibre of same type but different length, digital multimeter etc.

formula used :- i) $\alpha = \frac{20}{L} \log_{10}\left(\frac{V_1}{V_2}\right)$, $L = L_2 - L_1$

ii) $\alpha_p = \frac{\alpha}{4.343}$



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Experiment-6

Aim :- To determine the attenuation and propagation losses in optical fibre.

Apparatus Required :- He-Ne laser, 20× microscopic objective, fibre chuck, two optical fibre of same tube but of different length, digital multimeter, or photodetector.

formula used :- i) $\alpha = \frac{20}{L} \log_{10}\left(\frac{V_1}{V_2}\right)$ where $L = L_2 - L_1$

ii) $\alpha_p = \frac{\alpha}{4.343}$

where,

V_1 = voltage at output of fibre length of L_1

V_2 = " " " " " " L_2

α = fibre loss

α_p = attenuation efficient.

Theory :- As light travels along a fibre, its power decreases exponentially with distance. P at a distance inside a fibre

$P = P_0 e^{-\alpha_p z}$ — (i)

α_p = fibre attenuation efficient of unit $(km)^{-1}$. α called fibre loss

$\alpha = \frac{10}{z} \log_{10}\left(\frac{P_0}{P}\right)$ — (ii)

To find relation b/w α and α_p :-

$\log_{10} e = \frac{1}{2.303} \ln x$

Teacher's Signature

Observation and Calculation :-

i) Length of given optical fibre, $L = (L_2 - L_1) = 1m = 10^{-3} km$

ii) Voltage at output reading, $V_1 = 282\ mV$

iii) " " " , $V_2 = 205\ mV$

Fibre loss, $\alpha = \frac{20}{L} \log_{10}\left(\frac{V_1}{V_2}\right)$

$= \frac{20}{10^{-3}} \log_{10}\left(\frac{282}{205}\right) \Rightarrow 20000 \log_{10}(1.424)$

$= 20000 \times 0.1525$

$= 3070\ dB/km$

$\alpha_p = \frac{\alpha}{4.343} = \frac{3070}{4.343}$

$\alpha_p = 706.88\ km^{-1}$

#Result :- i) fibre loss is given by $\alpha = 3070\ dB/km$

ii) Attenuation coefficient is $\alpha_p = 706.88\ km^{-1}$

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Hence (ii) :- $\alpha = \frac{10}{(2.303)z} \ln\left(\frac{P_i}{P}\right) = \frac{4.343}{z} \ln\left(\frac{P_0}{P}\right)$

$\frac{\alpha z}{4.343} = \ln\left(\frac{P_0}{P}\right) \Rightarrow e^{\frac{\alpha z}{4.343}} = \frac{P_0}{P} \Rightarrow P = P_0 e^{-\frac{\alpha z}{4.343}}$ — (iii)

After comparing (i) & (iii) we get :-

$\frac{\alpha}{4.343} = \alpha_p \Rightarrow \alpha = 4.343\ \alpha_p$ — (iv)

then :-

$\alpha = \frac{10}{z} \log_{10}\left(\frac{P_0}{P_2}\right) \Rightarrow z = \frac{10}{\alpha} \log_{10}\left(\frac{P_0}{P_1}\right)$ — (v)

$\alpha = \frac{10}{z+L} \log_{10}\left(\frac{P_1}{P_2}\right) \Rightarrow z + L = \frac{10}{\alpha} \log_{10}\left(\frac{P_1}{P_2}\right)$ — (vi)

Substract (v) from (vi), we get

$L = \frac{10}{\alpha}\left[\log_{10}\left(\frac{P_0}{P_2}\right) - \log_{10}\left(\frac{P_0}{P_1}\right)\right] = \frac{10}{\alpha} \log_{10}\left(\frac{P_1}{P_2}\right)$ — (vii)

If R is the resistance of photodetector circuit, then electrical power is :-

$P_1 = \frac{V_1^2}{R}$ and $P_2 = \frac{V_2^2}{R}$

$\frac{P_1}{P_2} = \frac{V_1^2}{V_2^2} = \left(\frac{V_1}{V_2}\right)^2$, put this in eq^n (vii), we get.

$\alpha = \frac{10}{L} \log_{10}\left(\frac{V_1}{V_2}\right)^2$ or $\alpha = \frac{20}{L} \log_{10}\left(\frac{V_1}{V_2}\right)$ — (viii)

Since, measured using sensitive digital voltmeter, so eqn (viii) can be used to calculate fibre loss

Teacher's Signature

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Procedure :- (i) Couple the lesser output to a 20x microscopic objective then hold both ends of optical fibre of length, L inches.

(ii) Couple one end of fibre to objective and other end to photodetector. photodetector is itself connected to voltmeter.

(iii) Find $L = L_2 - L_1$ Put observed value in formula to find fibre loss.

(iv) perform experiment with glass as well as plastic fibres.

Precautions :- (i) coupling of light into fibre and into photodetector should be almost perfect.

(ii) Two fibres should be exactly similar. Only their length should be different.

(iii) There should be no sharp bends in dark room.

(iv) Light from output should fall only on photodetection.

Result :- (i) fibre loss is given by $\alpha = 3070$ dB/Km

(ii) Attenuation coefficient is $\alpha_p = 706.88$ km^{-1}

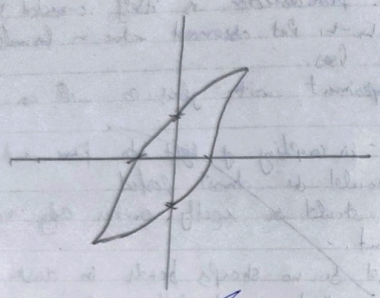
(iii) There is more power loss in plastic fibre as compared to glass fibre.

Teacher's Signature _____

Aim :- To study hysteresis loop for given ferromagnetic material on a CRO, using solenoid.

Apparatus :- CRO, ferromagnetic material, solenoid, hysteresis loop tracer.

Digram :-



Observation :-

Diameter of pick up coil = 3.21 mm

Length of sample, ℓ = 39 mm

diameter of sample, a = 1.17 mm

A_s = 1074

A_c = 8088

$\frac{A_s}{A_c}$ = 0.133 mm

Representation factor. $N = \ell/a = 33.33$ or $G_0 = 28.2$ G/V

B_x = 100 (Total Given Amplitude)

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Experiment 7

Aim :- To study hysteresis loop for given ferromagnetic material on a CRO using solenoid.

Apparatus :- CRO, ferromagnetic sample, solenoid, hysteresis loop

formula used :-

1. Coercitivity, $H_c = \frac{G_0 e_m}{\left(\frac{A_s}{A_c} - N\right)}$; $e_m = \frac{1}{2}$ loop width.

2. Saturation magnetization, $M_s = \frac{I}{4\pi}$, $\left[I = \frac{G_0 \mu_0 g_m (G_y)_s}{G_y \left(\frac{A_s}{A_c} - N\right)} \right]$

3. Retentivity, $M_R = \frac{I_R}{4\pi}$, $\left[I_R = \frac{G_0 \mu_0 g_m (e_y) s}{g_y \left(\frac{A_s}{A_c} - N\right)} \right]$

$(e_y)_s = \frac{1}{2}$ intercept.

Procedure :-

1. Take sample holder, insert a ferromagnetic sample, in lower pick.
2. Insert sample holder in solenoid, connect solenoid to solenoid socket of loop tracer using 8 pin cable.
3. Connect Y, X terminal of loop tracer to CRO Y, X terminal with help of BNC to 4 mm connector cable.

Teacher's Signature

Calculations :-

Loop width	Soft iron	Hard iron
Tip Tip Height	40	36
Intercept	32	26
Loop width	6	14

Putting these values in formula given :-

1. coercitivity, $H_c = \frac{G_0 \, e_n}{(A_y/A_x - N)}$

2. Saturation Magnetization, $M_s = \frac{B_s}{4\pi}$

3. Retentivity, $M_r = \frac{B_r}{4\pi}$

Result :-

Sample	Coercitivity	Saturation magnetization	Retentivity
Soft iron	650.720e	0.034 G	0.034 G
Hard iron	1517.20e	0.039 G	0.[illegible] G

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5. Connect +ve terminal of loop tracer to CRO, -ve terminal with help of BNC.
4. Keep CRO in XY mode.
5. After installation of complete set up, switch ON the Hysteresis loop tracer and CRO.
6. Adjust magnetic field intensity, magnetic field will be displayed on Gauss.
7. Now, Hysteresis loop of sample will be displayed on CRO.
8. Adjust the loop to read all parameters easily.

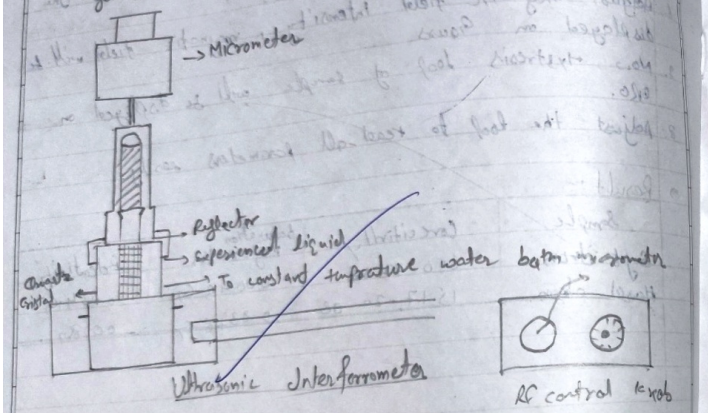
⇒ Result :-

Sample	Coercitivity	Saturation magnetization	Retentivity
Soft iron	650.72 oe	0.0043 G	0.034 G
Hard iron	1517.20 oe	0.039 G	0.038 G

Teacher's Signature

Aim :- To find the velocity of ultrasonic wave in a given liquid using ultrasonic interferences

Apparatus :- ultrasonic interferometer, sample liquid, high frequency generator.



formula used :- $v = \nu\lambda$

$\lambda = 2 \times$ loop width

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Experiment-8

Aim :- To find the velocity of ultrasound in a given liquid using ultrasonic interferometer.

Apparatus :- ~~If λ is wavelength of ultrasonic waves in a given liquid~~ Ultrasonic interferometer, sample liquid, high frequency generator.

formula used :- If λ is wavelength of ultrasonic waves in a given liquid, and d is distance between two consecutive ~~maxima~~ maxima, then velocity of ultrasonic waves is given by

$$v = 2d\nu$$

$$\lambda = 2 \times \text{loop width}$$

Theory :- The schematic interferometer, ultrasonic waves are produced using piezoelectric method. frequency of ultrasonic wave is equal to frequency of oscillation of quartz crystal.

Procedure :-

1) Fill the ultrasonic cell with given liquid.
2) Adjust frequency of R.F generator so that dielectric current is noted down.
3) Adjust micrometer screw slowly so that current shows minimum value; find the distance moved by reflecting rod reading.

Teacher's Signature

Observation :- x_1 = 1.38 mm, x_{11} = 4.79 mm

loop width = $\frac{x_{11} - x_1}{10}$ = 0.341 mm

Calculations :- $v = \nu \lambda$

$= 2 \times 10^6 \times 2 \times 0.341 \times 10^{-3}$

$v = 1366 \text{ ms}^{-1}$

Result :-

$v = 1366 \text{ ms}^{-1}$

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(v) Repeat the procedure multiple times.
(vi) Calculate value of λ and then velocity of ultrasound.
(vii) Note down frequency of ultrasonic from R.F generator then

Precautions :
1) Least count of screw gauge should be carefully noted
2) Screw must be moved in one direction.
3) ultrasonic cell should not be overfilled with liquid
4) There should not be any air gap b/w the quartz plate and reflecting plate.

Result :-

$V = 1366 \text{ ms}^{-1}$

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